

Coarsening Algorithmic Scores

Why can it be optimal to coarsen an already calibrated predictive score?

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I Why coarsen?

divided cost incidence $\rightarrow$ profitable local pooling

II Posterior-based costs

fixed stopping posteriors $\rightarrow$ affine review geometry $\rightarrow \leq 3$

III.A Experiment-based costs

finite signals $\rightarrow$ finite interface; continuous signals $\rightarrow$ residual disclosure can emerge

III.B Liability

more dispersed boundary targets $\rightarrow$ finite targeting becomes harder

IV Welfare

 $\rightarrow$ continuation value – assignment loss; redesign can overturn direct incentive gains

I Why coarsen a calibrated score?

Setting and timing

 $\theta \in \{\text{Good, Defective}\}, \omega \sim F, \Pr(\text{Good} \mid \omega) = \omega$
Accepting a Defective object costs d_A ; rejecting a Good object costs d_R . $a \in \{\text{Accept, Reject}\}, \ell(\mu) = \min\{\mu d_R, (1-\mu)d_A\}$
 $\bar{\mu} = d_A / (d_A + d_R)$ Reject below $\bar{\mu}$, accept above $\bar{\mu}$. Correct decisions have zero loss.The provider commits to reporting policy Γ before ω is realized. $\mu(m) = E[\omega \mid m], G \in \text{MPC}(F)$ Γ is the reporting policy; together with F it induces the posterior distribution G . Feasibility requires G to be a mean-preserving contraction of F : reporting can pool inherited score information but cannot create more.**Information boundary.** The decision maker observes report m , not the underlying score ω .

Control and objectives

At posterior μ , continuation x generates mistake loss $L_x(\mu)$ and resource cost $C_x(\mu)$. Let $L^*(\mu)$ and $C^*(\mu)$ denote the selected mistake loss and resource cost. The decision maker bears share α of mistake loss and pays the resource cost.

Provider

Before score realization

 $\max_{\Gamma} E_G[-(1-\alpha)L^*(\mu)]$
Chooses the reporting policy.

Decision maker

After report m $x^*(\mu) \in \arg \min_x \{\alpha L_x(\mu) + C_x(\mu)\}$
Chooses continuation and retains final-action authority.**Same mistake objective, different cost incidence:** the decision maker pays the resource cost; the provider bears residual mistake loss.

Disclosure force

 $V^H(\mu) := -\min_x \{\alpha L_x(\mu) + C_x(\mu)\}$ is convex.

Finer disclosure lets the decision maker tailor continuation more precisely.

Full disclosure is provider optimal exactly when $V^H + C^*$ is convex.

Costly-switch wedge

At continuation boundary μ_g , entering the more resource-intensive continuation raises resource cost by $\Delta C_g > 0$. Decision-maker indifference implies

$$L_g^- - L_g^+ = \Delta C_g / \alpha.$$

provider gain = surplus gain = $(1-\alpha)/\alpha \cdot \Delta C_g > 0$

A small asymmetric pool can move its common posterior to the high-resource side, inducing more resource-intensive continuation and fewer mistakes.

A local pool strictly raises provider payoff and surplus.

Near any costly continuation switch, an asymmetric pool can improve both objectives.

LOCAL $\neq$ GLOBAL. A local reason to pool does not determine global optimal reporting granularity.

II Posterior-Based Information Costs

Terminal-posterior choice and finite reporting

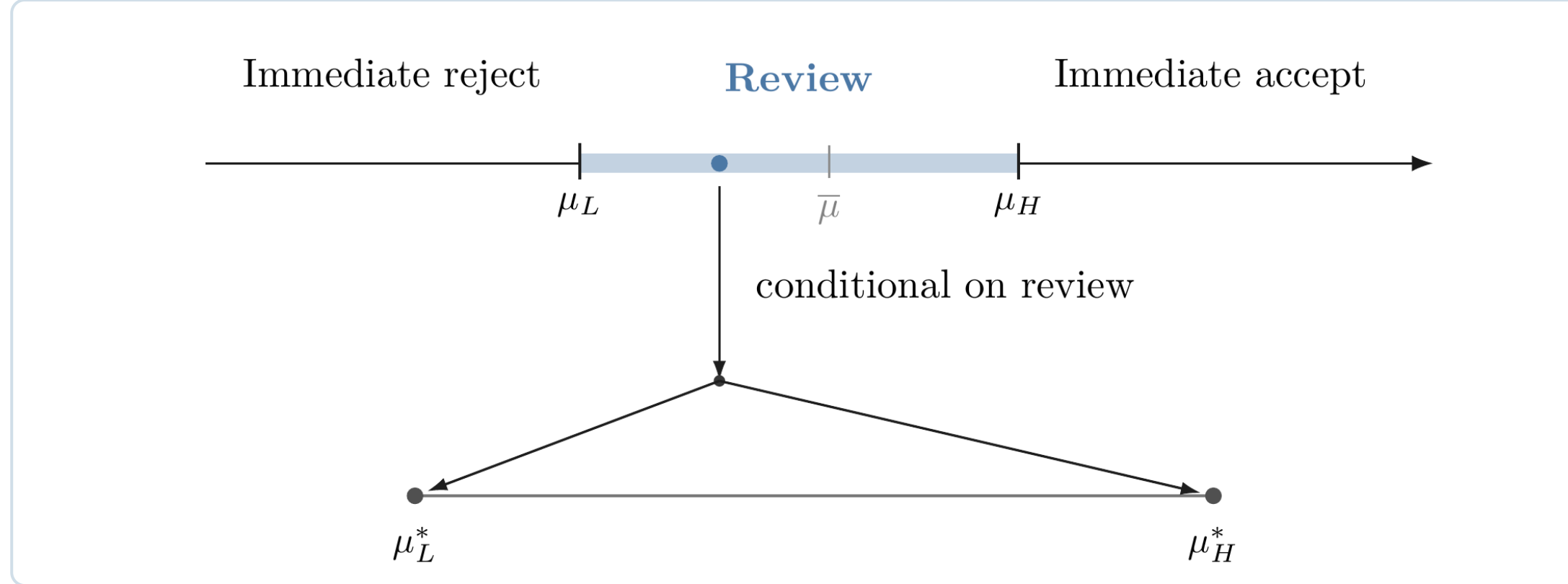
A. Choice object and cost

$$\tau \in \Delta([0,1]), E_{\tau}[v] = \mu$$
$$C(\tau; \mu) = 1\{\mathcal{J}(\tau; \mu) > 0\}K + \lambda \mathcal{J}(\tau; \mu)$$
$$\mathcal{J}(\tau; \mu) = \varphi(\mu) - E_{\tau}[\varphi(v)]$$

Posterior-based choice object. At each induced posterior μ , the decision maker reoptimizes over Bayes-plausible distributions τ of terminal posteriors with mean μ . Here v is a terminal posterior and φ is the common strictly concave potential. K is the separate activation cost; $\lambda \mathcal{J}$ is the variable information cost.

B. Conditional review

$$\tau^*(\mu) = q(\mu) \delta_{\mu_L^*} + [1-q(\mu)] \delta_{\mu_H^*}$$

 $q(\mu)$ is the mean-matching weight.**Fixed stopping posteriors.** Conditional review reaches the same two stopping posteriors throughout the review region. The mean-matching weights vary with the induced posterior, and implementing those same stopping posteriors generally requires the Blackwell experiment to change with μ . Review mistake loss is therefore affine in μ .fixed stopping posteriors $\rightarrow$ affine review loss $\rightarrow$ payoff-preserving score compression **$K = 0$: disclosure benchmark**

Review entry is continuous; gates coincide with stopping posteriors.

 $K > 0$, with review active: costly entry

Resource use jumps at review gates, making full disclosure strictly suboptimal.

A Finite Optimal Interface Exists

Posterior-based information costs

Some optimal reporting policy partitions the score support into at most three intervals.

One report is assigned to each interval.

Five supporting-price geometries behind the three-interval result

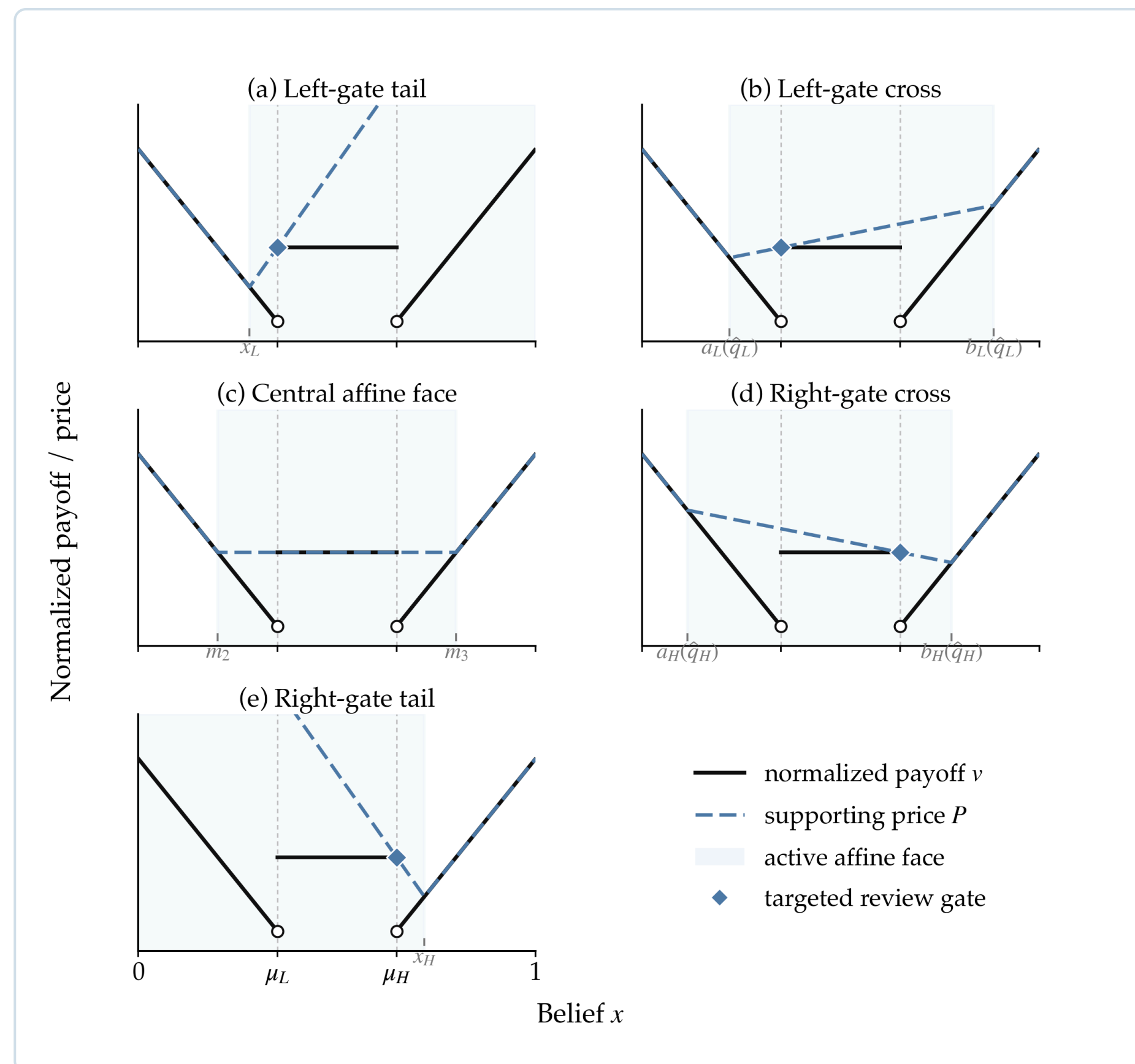


Figure key: black = normalized provider payoff · dashed blue = supporting price · shading = active affine face.

Five supporting-price geometries, two economic forms. Cases 1–2 and 4–5 are left/right one-gate targeting. Case 3 is the central affine-face design.

Each active affine face can be compressed without changing provider payoff.

III Experiment-Based Information Costs

Finite signals, continuous signals, and targeting

Experiment-based choice object. A given Blackwell experiment's state-contingent signal distributions and cost remain fixed as the induced posterior changes. The decision maker may still select different available experiments at different induced posteriors.

3.1 Finite-Signal Experiments and Finite Interfaces

$$L_{\mathcal{J}}(\mu) = \sum_{s \in \mathcal{S}_{\mathcal{J}}} \min\{d_R \mu p_G^j(s), d_A(1-\mu)p_D^j(s)\}$$

 $\mathcal{S}_{\mathcal{J}}$ is finite; $p_G^j(s)$ and $p_D^j(s)$ are experiment j 's state-contingent signal probabilities.finite signal support $\rightarrow L_{\mathcal{J}}(\mu)$ piecewise affine $\rightarrow$ provider payoff has a finite affine partition $\rightarrow$ some finite optimum exists**A finite menu of finite-signal experiments admits some optimal reporting interface with finitely many reports.** Sufficient, not necessary.

3.2 Continuous Signals and Residual Disclosure

continuously dispersed likelihood ratios $\rightarrow$ continuum of action-switch beliefs $\rightarrow L_e(\mu)$ strictly concave $\rightarrow$ active provider-payoff branch strictly convex

A fixed regular continuous-signal experiment can therefore generate payoff-relevant curvature that finite boundary targeting may fail to absorb.

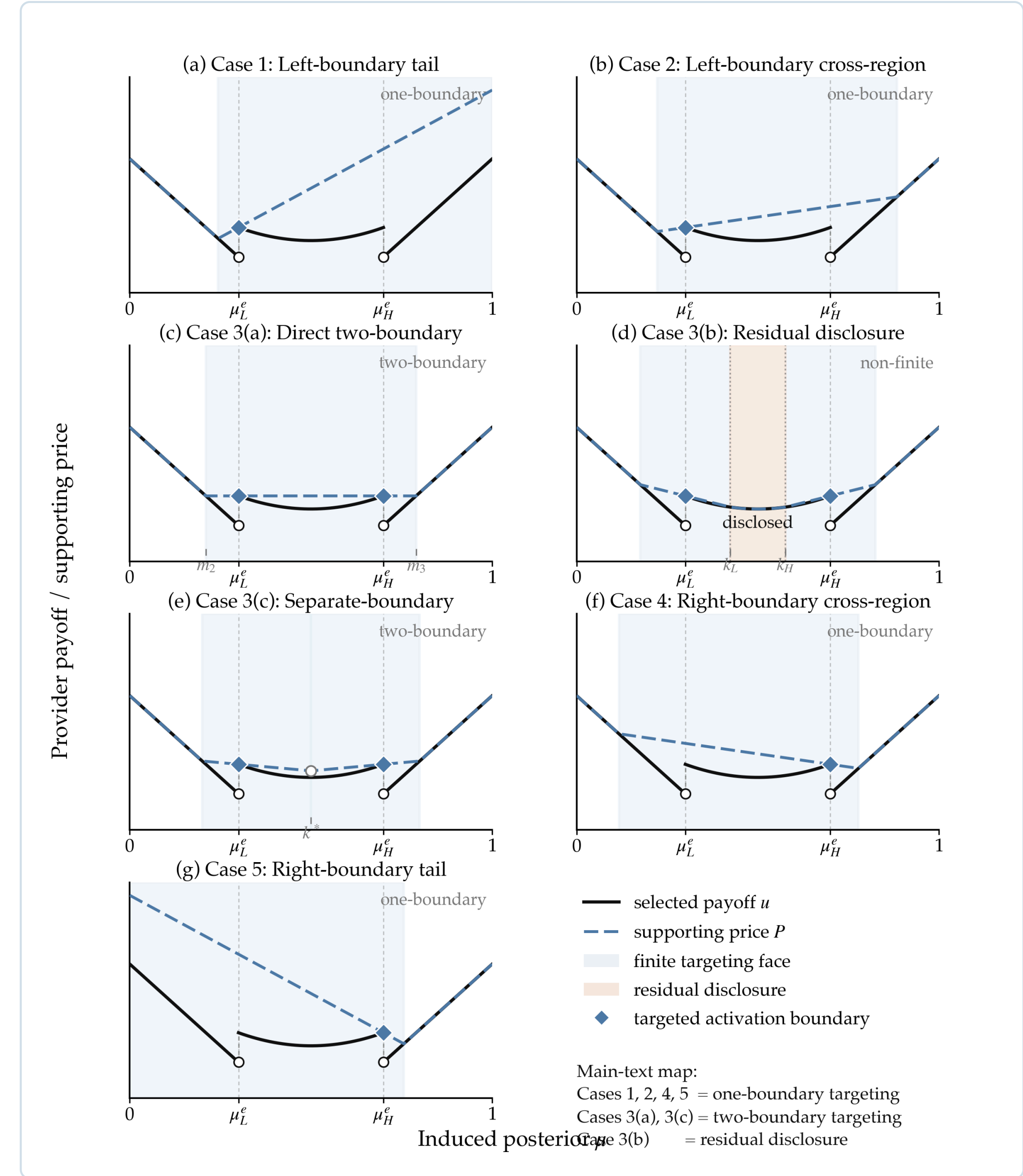


Figure key: black = provider payoff · dashed = supporting price · blue diamonds = activation boundaries · blue shading = finite targeting · orange = residual disclosure.

Seven geometries, three economic configurations. Cases 1–2 and 4–5 are one-boundary targeting. Case 3(a) is direct two-boundary targeting. Case 3(c) is separate-boundary targeting, and Case 3(b) is residual disclosure. The three central cases differ in whether the same two activation boundaries can be implemented by one central score block, by separately balanced score blocks, or not by any finite optimal interface.

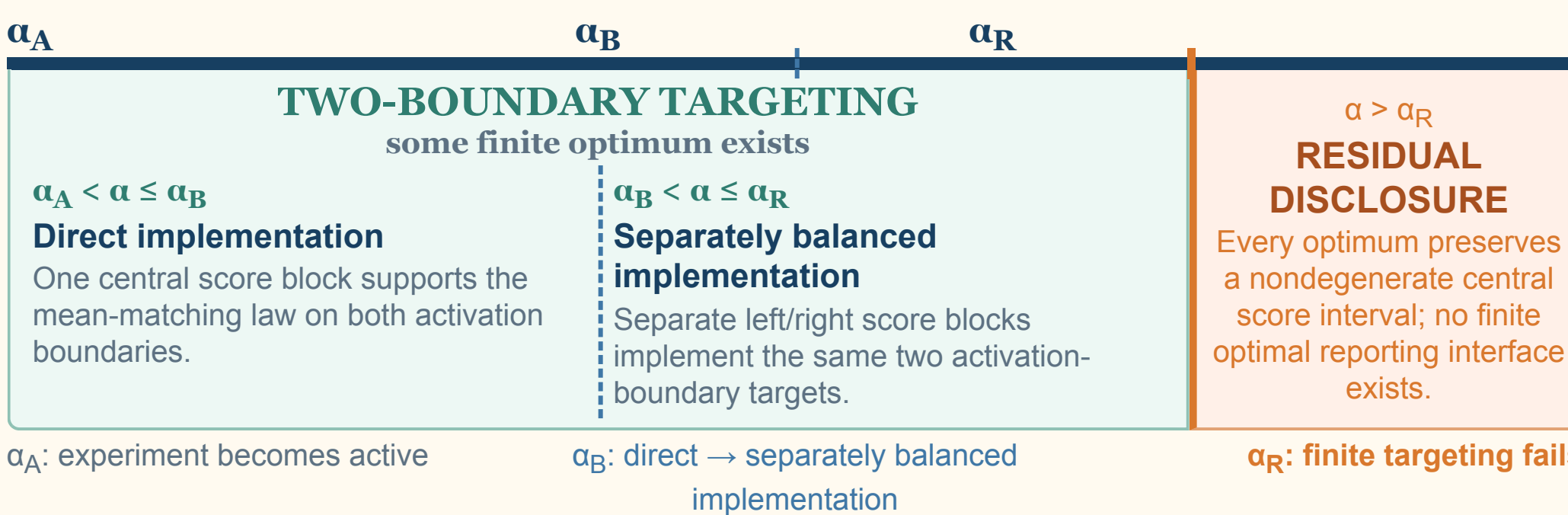
Liability changes the targeting problem

higher liability

acquisition region expands

activation boundaries move apart

target posteriors become more dispersed

Nonparametric symmetric class: $\alpha_A < \alpha_B \leq \alpha_R \leq 1$  α_A : experiment becomes active α_B : direct $\rightarrow$ separately balanced implementation α_R : finite targeting fails**Higher liability first changes how the same two boundaries are targeted. If $\alpha_R < 1$, finite targeting fails only at the higher threshold α_R .****Scope.** Comparisons are within active acquisition and $\alpha < 1$. The α_B implementation threshold follows from the continuous-signal primitives, symmetry, and the unimodal score density. The ordered transition through α_R additionally requires log-concavity of the experiment's gross value.

IV Welfare and Score Assignment

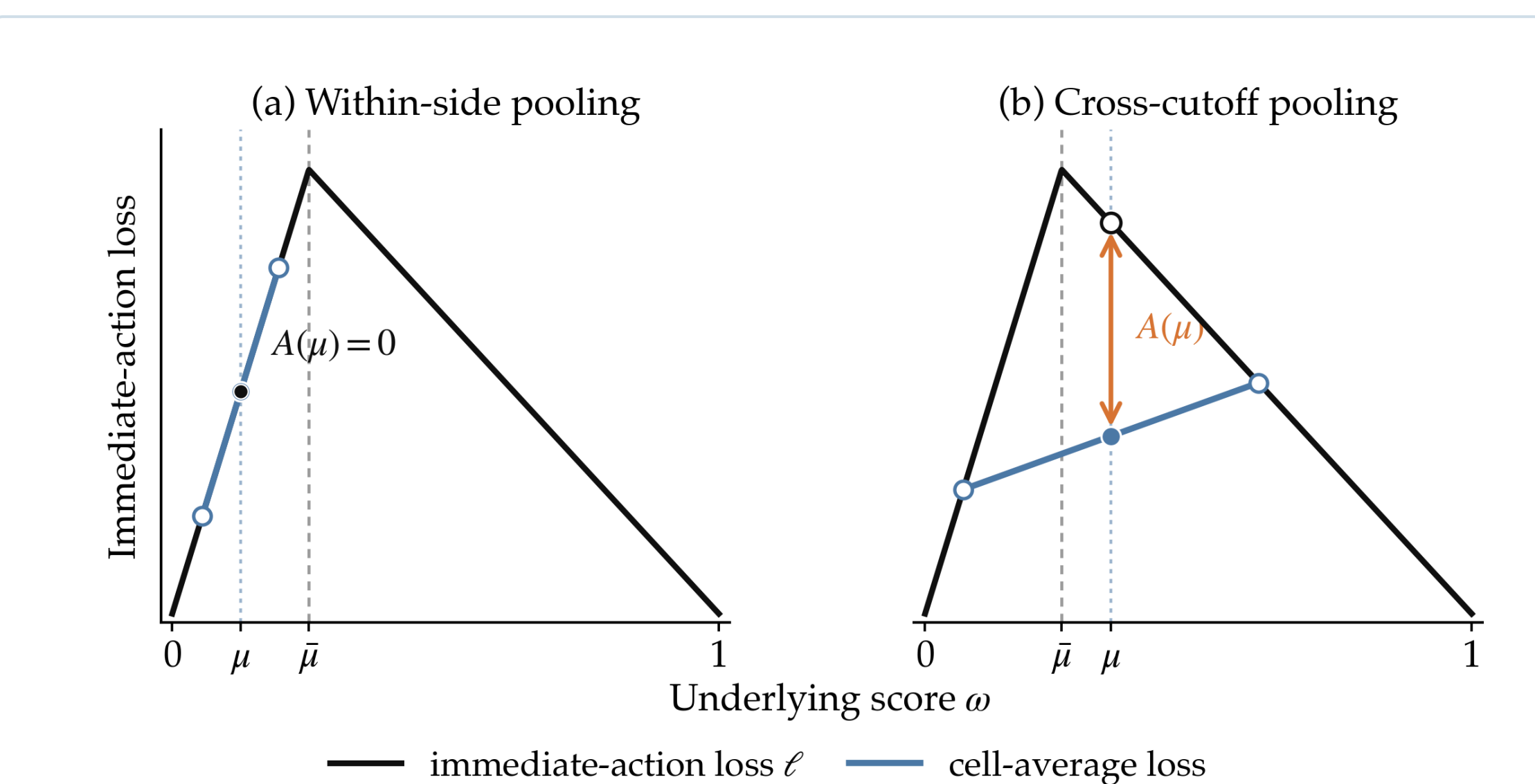
Continuation value, assignment loss, and liability redesign

Continuation value vs. score-assignment loss

 W_{score} is welfare under score-only immediate action. Continuation value is the reduction in total loss generated by the selected continuation relative to immediate action at posterior μ .

$$W(\Gamma; \alpha) - W_{\text{score}} = E[\text{continuation value}] - E[A(\mu)]$$

$$A(\mu) := \ell(\mu) - E[\ell(\omega) \mid \mu] \geq 0$$

Score-assignment loss. $A(\mu)$ measures the action-relevant score variation suppressed when heterogeneous scores are assigned to the same posterior μ .**Pooling scores that induce the same immediate action creates no score-assignment loss.** Pooling across the action cutoff suppresses decision-relevant variation.**Score-assignment loss $A(\mu)$.** Within-side pooling stays on one affine segment of immediate-action loss, so $A(\mu) = 0$. Cross-cutoff pooling creates the orange gap.

Liability: incentives, mistake loss, and redesign

A. Fixed reporting interface

$$\alpha_2 > \alpha_1 \Rightarrow W(\Gamma; \alpha_2) \geq W(\Gamma; \alpha_1)$$

Holding the reporting policy fixed, higher decision-maker liability weakly lowers selected mistake loss and weakly improves total continuation surplus.

B. After provider reoptimization

Provider-optimal expected mistake loss weakly falls with liability.

This mistake-loss monotonicity survives endogenous redesign of the reporting interface.

C. Welfare change

$$\text{continuation-incentive effect} + \text{interface-redesign effect} \geq 0$$

Any welfare decline from higher liability must come from endogenous interface redesign and must outweigh the direct incentive gain.

Accuracy still does not rank welfare. At a fixed liability share, provider-optimal reporting can weakly reduce expected mistake loss relative to full score disclosure followed by optimal continuation, yet yield lower welfare than score-only immediate action.

EXTENSIONS IN THE APPENDIX

common price + liability share $\rightarrow$ adoption $\rightarrow$ integration reveals type $\rightarrow$ personalized reporting**Market adoption and endogenous liability.** The provider posts a common price and liability share before adoption. Integration reveals deployment type, after which reporting can be personalized. Liability changes type-specific downstream burdens, so common pricing creates a rent-compression motive.**optimal reporting $\rightarrow$ imperfect enforcement $\rightarrow$ credibility****Imperfect enforcement and credibility.** Under full commitment, the main reporting problem is unchanged. With case-by-case discretion to relabel reports, review-inducing reports may require sufficient protocol enforcement. This implementation extension does not alter the full-commitment optimal-design results.